

Khulna University of Engineering & Technology (KUET)

Department of Computer Science and Engineering

CLO–Assessment Mapping & Rubric-Based Evaluation of CSE 3200: System Development Project

Course Contents:

Students will work in groups or individually to develop a term project to solve complex engineering problems. It may include I/O drivers, operating systems modules, or software solutions of some practical problems involving complex engineering process. Students will develop a software system using recent technologies/programming maintaining software engineering principles with proper documentation.

Course Learning Outcomes (CLOs)

After completing this course, students will be able to:

- 1. CLO1: Problem Analysis:** Identify, formulate, and analyze complex engineering problems to define appropriate system requirements. *(Aligned with analytical and critical thinking skills)*
- 2. CLO2: System Design & Development:** Design and develop a complete software/system solution (e.g., OS module, I/O driver, or real-world application) using appropriate methodologies and modern technologies. *(Core engineering design competency)*
- 3. CLO3: Application of Modern Tools:** Select and apply suitable modern tools, programming languages, frameworks, and development environments for system implementation. *(Practical and technical skill development)*
- 4. CLO4: Investigation & Testing:** Conduct testing, validation, and performance evaluation of the developed system to ensure correctness, reliability, and efficiency. *(Experimental and investigative capability)*
- 5. CLO5: Teamwork & Project Management:** Work effectively as an individual or in a team to plan, manage, and execute a system development project following software engineering principles. *(Leadership, collaboration, and management skills)*
- 6. CLO6: Communication & Documentation:** Prepare comprehensive technical documentation and present the project outcomes effectively to stakeholders. *(Professional communication skills)*

CLO–PLO Mapping (BAETE / Washington Accord Standard)

Below is a suggested mapping (✓ = strong correlation, ○ = moderate correlation):

CLO \ PLO	PLO ₁	PLO ₂	PLO ₃	PLO ₄	PLO ₅	PLO ₆	PLO ₇	PLO ₈	PLO ₉	PLO ₁₀	PLO ₁₁	PLO ₁₂
CLO ₁	✓	✓	○	○								○
CLO ₂	✓	✓	✓	○	✓	○			○		○	○
CLO ₃	○		○		✓							✓
CLO ₄	○	✓	○	✓	✓							○
CLO ₅			○		○			✓	✓		✓	○
CLO ₆									○	✓		○

Notes for BAETE Compliance

- ✓ Ensure each CLO is **measurable and assessable** (use action verbs from Bloom's Taxonomy).
- ✓ Coverage of **PLO₃, PLO₅, PLO₉, PLO₁₀, PLO₁₁** is essential for project-based courses.
- ✓ Include **rubrics** for:
 - Design quality
 - Implementation
 - Testing
 - Documentation
 - Presentation

1. CLO–Assessment Mapping with Marks Distribution

Assessment Tool	Description	Marks
A1: Proposal & Problem Analysis	Problem identification, literature review, feasibility	10
A2: System Design	Architecture, methodology, design diagrams	15
A3: Implementation & Tool Usage	Coding, tools, technologies	20
A4: Testing & Evaluation	Verification, validation, performance	15
A5: Documentation	Report writing (SRS, design, user manual)	15
A6: Presentation & Viva	Oral presentation, demo, defense	15
A7: Teamwork & Project Management	Planning, task distribution, progress tracking	10
Total		100

2. CLO–Assessment Mapping Matrix

CLOs ↓ / Assessment →	A1 (10)	A2 (15)	A3 (20)	A4 (15)	A5 (15)	A6 (15)	A7 (10)	Total Contribution
CLO1: Problem Analysis	✓ (10)							10
CLO2: Design & Development		✓ (10)	✓ (10)					20
CLO3: Modern Tools			✓ (10)					10
CLO4: Testing & Investigation				✓ (15)				15
CLO5: Teamwork & Management		✓ (5)					✓ (10)	15
CLO6: Communication					✓ (15)	✓ (15)		30

3. Detailed Rubrics for Each CLO (BAETE)

Performance Levels:

Level	Description	Score Range
Excellent (4)	Exceeds expectations with strong depth and quality	80–100%
Good (3)	Meets expectations with minor limitations	65–79%
Satisfactory (2)	Basic achievement, lacks depth	50–64%
Poor (1)	Fails to meet minimum requirements	<50%

CLO1: Problem Analysis (10 Marks)

Criteria	4 (Excellent)	3 (Good)	2 (Satisfactory)	1 (Poor)
Problem Definition	Clear, precise, well-scoped	Mostly clear	Vague or partially defined	Unclear
Requirement Analysis	Complete & justified	Mostly complete	Some missing	Incomplete
Feasibility Study	Comprehensive analysis	Moderate	Limited	Not done

CLO2: System Design & Development (20 Marks)

Criteria	4 (Excellent)	3 (Good)	2 (Satisfactory)	1 (Poor)
System Architecture	Well-structured, scalable	Good design	Basic structure	Poor design
Algorithm/Logic	Efficient and optimized	Mostly correct	Basic logic	Incorrect
Design Tools (UML, etc.)	Complete & professional	Mostly complete	Partial	Missing

CLO3: Modern Tools Usage (10 Marks)

Criteria	4 (Excellent)	3 (Good)	2 (Satisfactory)	1 (Poor)
Tool Selection	Highly appropriate	Appropriate	Limited	Inappropriate
Tool Application	Expert use	Good use	Basic use	Poor use
Technology Integration	Seamless integration	Minor issues	Partial	Not integrated

CLO4: Testing & Investigation (15 Marks)

Criteria	4 (Excellent)	3 (Good)	2 (Satisfactory)	1 (Poor)
Test Case Design	Comprehensive	Adequate	Limited	Poor
Validation & Verification	Thorough	Moderate	Basic	Missing
Performance Analysis	Detailed insights	Some analysis	Minimal	None

CLO5: Teamwork & Project Management (15 Marks)

Criteria	4 (Excellent)	3 (Good)	2 (Satisfactory)	1 (Poor)
Team Contribution	Highly collaborative	Mostly collaborative	Limited	Poor
Task Management	Well planned & tracked	Adequate	Some planning	No planning
Responsibility & Leadership	Strong leadership	Good	Basic	Weak

CLO6: Communication & Documentation (30 Marks)

Criteria	4 (Excellent)	3 (Good)	2 (Satisfactory)	1 (Poor)
Report Quality	Professional, well-structured	Good	Basic	Poor

Criteria	4 (Excellent)	3 (Good)	2 (Satisfactory)	1 (Poor)
Technical Content	Comprehensive	Adequate	Limited	Weak
Presentation Skills	Clear, confident, engaging	Good	Moderate	Poor
Q&A / Defense	Excellent responses	Good	Basic	Unable

4. Notes for Implementation (BAETE Compliance)

- ✓ Use **direct assessment tools** (presentation, report, demo) for CLO attainment.
 - ✓ Maintain **individual contribution records** for fairness.
 - ✓ Apply **rubric-based scoring** to ensure transparency.
 - ✓ Convert marks → **CLO attainment levels** (e.g., $\geq 60\%$ = attained).
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Example Scenario

- 1) **Project Title:** Smart Traffic Monitoring System
- 2) **Group:** G1 (3 students)
- 3) **Student Evaluated:** Student A
- 4) **Total Marks:** 100

1. Assessment Scores (Raw Marks Given by Instructor)

Assessment	Max Marks	Obtained Marks
A1: Proposal & Problem Analysis	10	8
A2: System Design	15	12
A3: Implementation	20	16
A4: Testing	15	11
A5: Documentation	15	13
A6: Presentation & Viva	15	12
A7: Teamwork & Management	10	9
Total	100	81

2. CLO-wise Attainment Calculation

CLO₁ (Problem Analysis) → A₁

- Score = $8/10 = 80\%$ → **Level 4 (Excellent)**

CLO₂ (Design & Development) → A₂ + A₃

- A2 contribution = 12/15
- A3 contribution (design part) = 8/10 (assumed split)
- Total = (12 + 8) / 20 = **20/20 = 100% → Level 4**

CLO₃ (Modern Tools) → A₃

- Tool-related marks = 8/10
- Score = 8/10 = **80% → Level 4**

CLO₄ (Testing & Investigation) → A₄

- Score = 11/15 = **73% → Level 3 (Good)**

CLO₅ (Teamwork & Management) → A₂ + A₇

- A2 teamwork portion = 4/5
- A7 = 9/10
- Total = (4 + 9) / 15 = **13/15 = 86% → Level 4**

CLO₆ (Communication) → A₅ + A₆

- A5 = 13/15
- A6 = 12/15
- Total = 25/30 = **83% → Level 4**

3. Summary of CLO Attainment

CLO Marks Percentage Attainment Level

CLO ₁	8/10	80%	4 (Excellent)
CLO ₂	20/20	100%	4 (Excellent)
CLO ₃	8/10	80%	4 (Excellent)
CLO ₄	11/15	73%	3 (Good)
CLO ₅	13/15	86%	4 (Excellent)
CLO ₆	25/30	83%	4 (Excellent)

4. Interpretation (BAETE)

- **All CLOs are attained (≥60%) ✓**
- Strong performance in:
 - Design & Development (CLO₂)
 - Communication (CLO₆)
- Slight improvement needed in:

- Testing & validation (CLO₄)

5. Rubric-Based Justification Example

Example for CLO₄ (Testing):

- Test cases were present but not exhaustive → **Level 3**
- Performance analysis was moderate → **Level 3**
- Hence, overall rubric level = **Good (3)**

6. Optional: Class-Level Attainment Example

If 30 students are evaluated:

CLO	Students ≥60%	Attainment (%)	Status
CLO ₁	26/30	87%	Attained
CLO ₂	28/30	93%	Attained
CLO ₃	25/30	83%	Attained
CLO ₄	22/30	73%	Attained
CLO ₅	27/30	90%	Attained
CLO ₆	29/30	97%	Attained

7. Summary

- Use **rubrics** → **marks** → **CLO mapping** → **attainment %**
- Maintain:
 - Individual evaluation sheets
 - Evidence (report, code, presentation slides)
- CLO attainment threshold (BAETE standard):
Typically **≥60% students scoring ≥60% marks**